

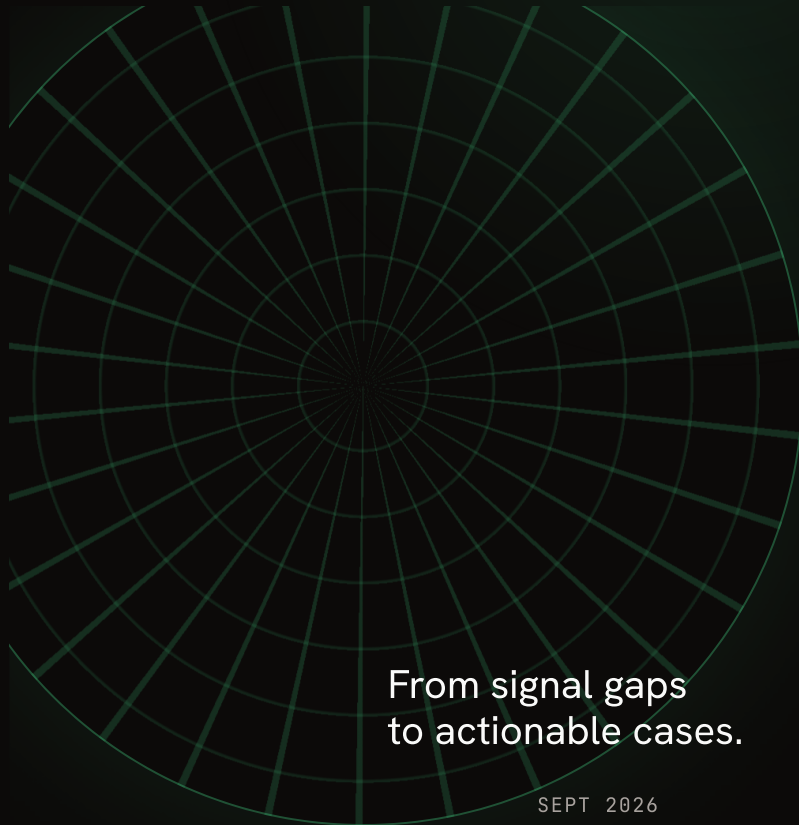
● MARITIME RISK INTELLIGENCE



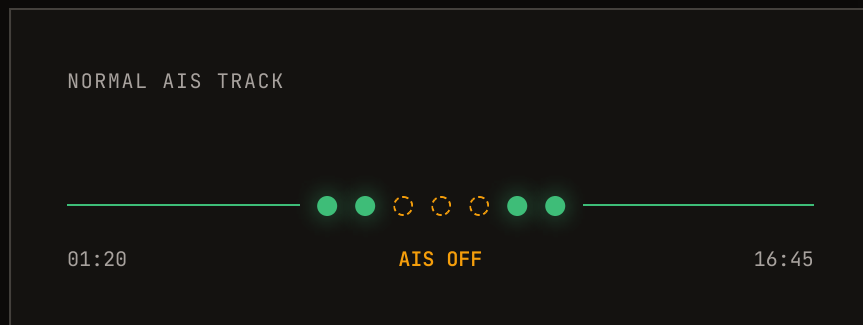
Wake AI

Making dark shipping
visible.

PENN BUILT



When a vessel goes dark, the trail goes cold.



Vessels can disable AIS, the system that broadcasts their location, speed, and heading.

That blackout can hide sanctioned oil movements and ship-to-ship cargo transfers.

Not a fringe problem.

558

average suspected
dark tankers

~25%

of the global crude
tanker fleet

7.8M

metric tons of crude
moved per month

43%

of officially recorded global seaborne crude exports in the study comparison

Source: Fernandez-Villaverde, Li, Xu & Zanetti (2025), "Charting the Uncharted," Oxford Discussion Paper 1070.

OXFORD / PENN RESEARCH - 2017-2023 ESTIMATES

One signal is never enough.

01

AIS gaps

When and where a vessel disappears

02

Geography

Risky routes and sanctioned-port proximity

03

Behavior

Speed, heading and voyage anomalies

04

Proximity

Potential ship-to-ship transfers

KEY PRINCIPLE An AIS gap is not proof. Context turns a gap into a lead.

Research context: Welch et al. (2022), "Hot Spots of Unseen Fishing Vessels"; Miller et al. (2018), "Identifying Global Patterns of Transshipment Behavior."

FRAGMENTED SIGNALS -> EXPLAINABLE RISK

Wake AI finds the signals worth waking up to.

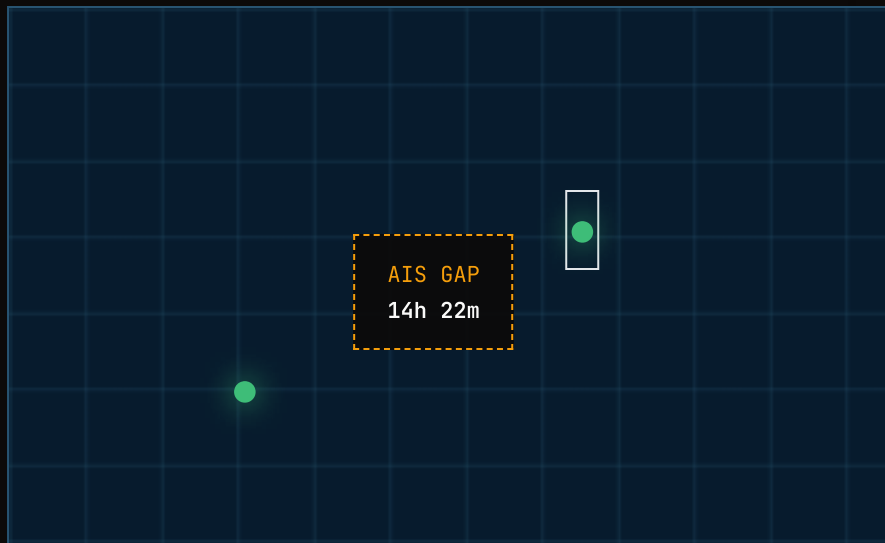


× DETECT × CONNECT × EXPLAIN × PRIORITIZE

Method context: [Welch et al. \(2022\)](#) and [Miller et al. \(2018\)](#).

FROM RAW SIGNALS TO INVESTIGATOR-READY CASES

Watch a blackout become a case.



Illustrative workflow; dark-shipping context from [Fernandez-Villaverde et al. \(2025\)](#).

LIVE PRODUCT WALKTHROUGH

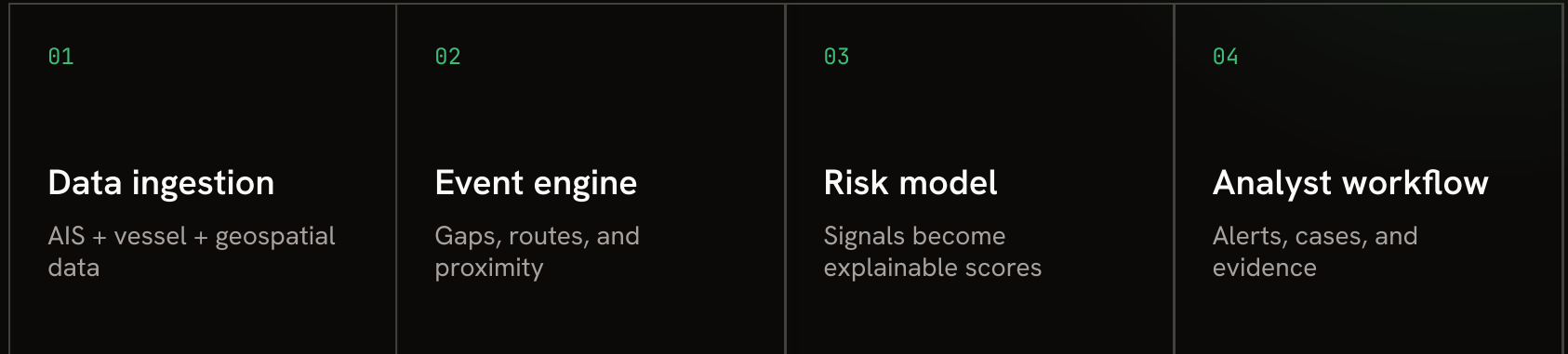
01 Normal voyage begins

02 AIS signal disappears

03 Wake AI finds correlated risk signals

04 Analyst receives the evidence-backed case

How Wake AI works.



DETAIL TO BE ADDED: data sources, model logic, and analyst interface.

Research context: Welch et al. (2022); Miller et al. (2018).

• WAKE AI

When ships go dark, risk should not.

Make maritime activity visible, explainable, and actionable.
